The Congruent Number Problem

Sagnik Dutta

Advised by Prof. Somnath Jha

Background and Motivation

Definition

A positive integer is called n is called a *congruent* number if it is area of a rational right triangle .

Background and Motivation

Definition

A positive integer is called n is called a *congruent* number if it is area of a rational right triangle .

- Congruent numbers have been in the topic of interest from 10th century by the arab mathematicians.
- A number n is congruent if and only if any $\lambda^2 n$ is also congruent for any non zero scalar λ so we will only consider square free positive integers when dealing with the congruent number .

Main Problem

- *Question*: Given a positive integer n, whether there is an algorithm to determine that number is congruent

Main Problem

- *Question*: Given a positive integer n, whether there is an algorithm to determine that number is congruent
- Any primitive pythagorean triplet is of the form $(m^2 - n^2, 2mn, m^2 + n^2)$ with $(m, n) = 1$ and $m > n$ $m \not\equiv n(mod 2)$, so listing out all the $mn(m^2 - n^2)$ being the area of such triangles will give us all the congruent numbers .

Main Problem

- *Question*: Given a positive integer n, whether there is an algorithm to determine that number is congruent
- Any primitive pythagorean triplet is of the form $(m^2 - n^2, 2mn, m^2 + n^2)$ with $(m, n) = 1$ and $m > n$ $m \not\equiv n(mod 2)$, so listing out all the $mn(m^2 - n^2)$ being the area of such triangles will give us all the congruent numbers .
- But in reality , this is not an efficient algorithm , as one will not list out all the integers in a finite amount of time

Table of congruent numbers

Following is a list of primitive triplets with $m + n \leq 9$. But this is not a good algorithm it is not systematic in appearence of the last column. So we can't say when a number will occur , if at all by building the table.

m	n	(a,b,c)	$\frac{1}{2}ab$	squarefree part
2	1	(3,4,5)	6	6
4	1	(15,8,17)	60	15
3	2	(5,12,13)	30	30
6	1	(35,12,37)	210	210
5	2	(21,20,29)	210	210
4	3	(7,24,25)	84	21
8	1	(63,16,65)	504	126
7	2	(42,28,53)	630	70
5	4	(9,40,41)	180	5

Examples

- 53 is *congruent* , but it comes when $m = 1873180325, n = 1158313156$(The correspodning area is $53 \times 297855654284978790^2$)

Examples

- 53 is *congruent* , but it comes when $m = 1873180325, n = 1158313156$(The correspodning area is $53 \times 297855654284978790^2$)
- $1, 2, 3$ are not *congruent*.(Fermat showed 1 is not *congruent* using the method of infinite descent.)

Examples

- 53 is *congruent* , but it comes when $m = 1873180325, n = 1158313156$(The correspodning area is $53 \times 29785565428497879 0^2$)
- $1, 2, 3$ are not *congruent*.(Fermat showed 1 is not *congruent* using the method of infinite descent.)
- 157 is a *congruent* with

$$x = \frac{6803298487826435051217540}{411340519227716149383203},$$

$$y = \frac{411340519227716149383203}{216665556937147613096 10},$$

$$z = \frac{224403517704336969924557513090674863160948472041}{89123322689288595880255351789671635700164808 30}.$$

Connection with Elliptic Curves

Suppose $n = \frac{1}{2}ab$ is a congruent number with $a^2 + b^2 = c^2$;
We get that;

$$(a+b)^2 = (\frac{c}{2})^2 + n \tag{1}$$

$$(a-b)^2 = (\frac{c}{2})^2 - n \tag{2}$$

Multiplying them together , and replacing $u = \frac{a^2-b^2}{2}; v = \frac{c}{2};$

$$u^2 = v^4 - n^2 \tag{3}$$

Replace $y = uv$ and $x = v^2$ we finally get ;

$$\mathbf{y^2 = x^3 - n^2 x} \tag{4}$$

which we denote by $\mathbf{E_n}$.

Connection with Elliptic Curves Ctd.

- n corresponds to a rational point on E_n whose x- coordinate is square of a rational number and also has even denominator.

Connection with Elliptic Curves Ctd.

- n corresponds to a rational point on E_n whose x- coordinate is square of a rational number and also has even denominator.
- Next natural question one can ask is whether any $(x, y) \in \mathbb{Q} \times \mathbb{Q}$ corresponds to a right triangle with area n in the above way .

Connection with Elliptic Curves Ctd.

- n corresponds to a rational point on E_n whose x- coordinate is square of a rational number and also has even denominator.
- Next natural question one can ask is whether any $(x, y) \in \mathbb{Q} \times \mathbb{Q}$ corresponds to a right triangle with area n in the above way .
- This is certainly not true as the x coordinate of that point must be a square of a some rational number and has even denominator .

Connection with Elliptic Curves Ctd.

- n corresponds to a rational point on E_n whose x- coordinate is square of a rational number and also has even denominator.
- Next natural question one can ask is whether any $(x, y) \in \mathbb{Q} \times \mathbb{Q}$ corresponds to a right triangle with area n in the above way .
- This is certainly not true as the x coordinate of that point must be a square of a some rational number and has even denominator .
- *Sufficient Condition :* Let $(x, y) \in \mathbb{Q} \times \mathbb{Q}$ on E_n such that (i) $x \in (\mathbb{Q}^+)^2$ (ii)x has even denominator ; then it corresponds to a rational right triangle having area n.

Group Law of Elliptic Curve

- The rational points on an elliptic curve can be made a group infact an abelian group as follows :
  Group Law: Let P, Q be two points on E.Then take the line joining them $\overline{PQ}$ (or the tangent line if $P = Q$).Let this line intersect at a third point say R.Denote $P \oplus Q$ to be reflection of R wrt to the x axis.

Group Law of Elliptic Curve

- The rational points on an elliptic curve can be made a group infact an abelian group as follows :
  Group Law: Let P, Q be two points on E.Then take the line joining them $\overline{PQ}$ (or the tangent line if $P = Q$).Let this line intersect at a third point say R.Denote $P \oplus Q$ to be reflection of R wrt to the x axis.
- (i) $(P \oplus Q) \oplus R = O$.
  (ii)$P \oplus Q = Q \oplus P$.
  (iii)For every P on E there is a $-P$ on E such that $P \oplus (-P) = O$.
  (iv)$(P \oplus Q) \oplus R = P \oplus (Q \oplus R)$.

Group Law of Elliptic Curve Ctd.

- $\mathbb{Q}$ points on an elliptic curve E i.e. $E(\mathbb{Q})$ is an abelian group.In fact Mordell showed that this is a finitely generated abelian group.

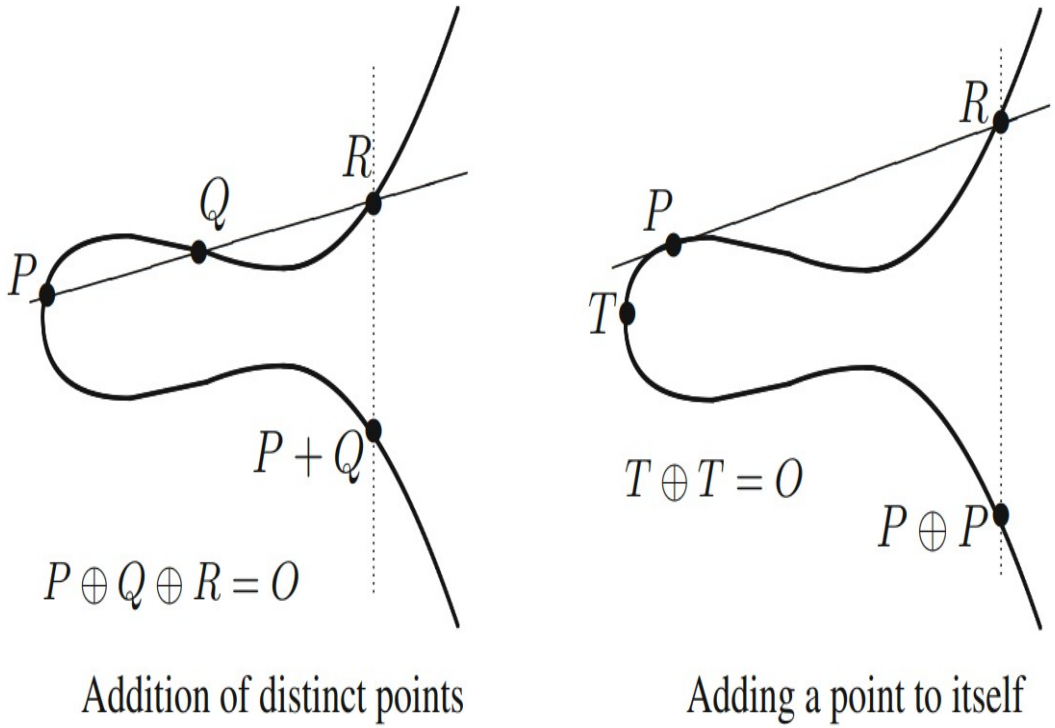


Figure: Addition Law of elliptic curves

Our Elliptic Curve E_n

Theorem

A positive integer n is congruent if and only if the corresponding E_n has non zero rank .

Our Elliptic Curve E_n

Theorem

A positive integer n is congruent if and only if the corresponding E_n has non zero rank .

Proof : By structure theorem of finitely generated abelian groups ;

$$E_n(\mathbb{Q}) \cong \mathbb{Z}^r \oplus E_n^{Tor}$$

r is the rank, here $E_n^{Tor} = \{(0,0), (\pm n, 0), O\}$.
Firstly if n is congruent number then we get a point (x, y) from $\mathbb{Q} \times \mathbb{Q}$ on E_n with $x \in (\mathbb{Q}^+)^2$, and thus that is not a torsion point showing E_n has non zero rank.
Conversely, if P is a point of infinite order then $x(2P)$ is square of a rational number and also it has even denominator,so we will get a rational right triangle having area n.

Hasse-Weil L Function

- The Hasse-Weil L function for the elliptic curve E_n

$$L(E_n, s) := \frac{\zeta(s)\zeta(1-s)}{\prod_p Z(E_n/\mathbb{F}_p; p^{-s})}; Re(s) > \frac{3}{2}$$

where $Z(E_n/\mathbb{F}_p, T) = exp(\sum_{r\geq 1} \#E(\mathbb{F}_{p^r})\frac{T^r}{r})$ and $\zeta(s) = \sum_{n=1}^{\infty} n^{-s}$ is the Riemann Zeta function .

Hasse-Weil L Function

- The Hasse-Weil L function for the elliptic curve E_n

$$L(E_n, s) := \frac{\zeta(s)\zeta(1-s)}{\prod_p Z(E_n/\mathbb{F}_p; p^{-s})}; Re(s) > \frac{3}{2}$$

  where $Z(E_n/\mathbb{F}_p, T) = exp(\sum_{r\geq 1} \#E(\mathbb{F}_{p^r})\frac{T^r}{r})$ and $\zeta(s) = \sum_{n=1}^{\infty} n^{-s}$ is the Riemann Zeta function .
- It has an Euler product $\prod_{p\nmid 2n} \frac{1}{1-2a_{n,p}p^{-s}+p^{1-2s}}$

Hasse-Weil L Function

- The Hasse-Weil L function for the elliptic curve E_n

$$L(E_n, s) := \frac{\zeta(s)\zeta(1-s)}{\prod_p Z(E_n/\mathbb{F}_p; p^{-s})}; Re(s) > \frac{3}{2}$$

  where $Z(E_n/\mathbb{F}_p, T) = exp(\sum_{r\geq 1} \#E(\mathbb{F}_{p^r})\frac{T^r}{r})$ and $\zeta(s) = \sum_{n=1}^{\infty} n^{-s}$ is the Riemann Zeta function .
- It has an Euler product $\prod_{p\nmid 2n} \frac{1}{1-2a_{n,p}p^{-s}+p^{1-2s}}$
- For each prime p if we expand each term in the product as geometric series and then multiply them together we will get a Dirichlet Series $L(E_n, s) = \sum_{m=1}^{\infty} b_{m,n}m^{-s}$

Main Theorem

- **Theorem:** The Hasse-Weil L function $L(E_n, s)$ extends analytically to an entire function on $\mathbb{C}$. Moreover , if

$$\Lambda(s) = (\frac{\sqrt{N}}{2\pi})^s \Gamma(s) L(E_n, s);$$

where $N = 32n^2 (resp. 16n^2)$ for n odd(resp. even);
$\Gamma(s) = \int_0^\infty exp(-t)t^{s-1}dt$
then it satisfies the functional equation

$$\Lambda(s) = C\Lambda(2-s)$$

with $C = (\frac{-2}{n})$ (resp. $(\frac{-2}{\frac{n}{2}})$) for n odd(resp. even) where for $n \equiv 5, 6, 7 (\text{mod } 8)$; $C = -1$, putting $s = 1$ then implies that $L(E_n, 1) = 0$.

Congruent Number Problem and BSD Conjecture

- The value at $s = 1$ of the L function is called the critical value and it has been conjectured that
  (Weak BSD Conjecture) An elliptic curve E has infinitely many rational points if and only if $L(E, 1) = 0$.

Congruent Number Problem and BSD Conjecture

- The value at $s = 1$ of the L function is called the critical value and it has been conjectured that
  (Weak BSD Conjecture) An elliptic curve E has infinitely many rational points if and only if $L(E, 1) = 0$.
- One side of the conjecture has already been proved for a large class of elliptic curves in which our elliptic curve E_n also belongs to.
  Coates and Wiles: If E is an elliptic curve defined over $\mathbb{Q}$ that has complex multiplication , then $L(E, 1) = 0$ if it has infinitely many rational points .

Congruent Number Problem and BSD Conjecture

- The value at $s = 1$ of the L function is called the critical value and it has been conjectured that
  (Weak BSD Conjecture) An elliptic curve E has infinitely many rational points if and only if $L(E, 1) = 0$.
- One side of the conjecture has already been proved for a large class of elliptic curves in which our elliptic curve E_n also belongs to.
  Coates and Wiles: If E is an elliptic curve defined over $\mathbb{Q}$ that has complex multiplication , then $L(E, 1) = 0$ if it has infinitely many rational points .
- So once we assume the BSD Conjecture , then from the last theorem it follows that
  If $\mathbf{n \equiv 5, 6, 7 (\mathrm{mod}\ 8)}$, then n is congruent.

Tunnell's Result

Suppose n be a square free odd natural number.
Then n is said to be congruent if and only if

$$\begin{aligned}\#\{(x,y,z) \in \mathbb{Z}^3 : 2x^2 + y^2 + 32z^2 = n\} \\ = \frac{1}{2}\#\{(x,y,z) \in \mathbb{Z}^3 : 2x^2 + y^2 + 8z^2 = n\} \quad (5)\end{aligned}$$

Thank You !